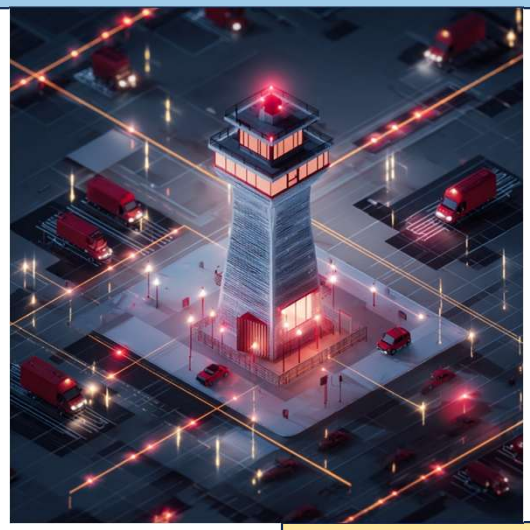
DSAI
Capstone Project
1: Ninja Van AI Control Tower

A Multi-Agent System for Predictive, Explainable, and Coordinated Logistics Decision-Making



2: Business Problem
3: Selected 5 Problems
4: Solution Overview
5: Architecture Diagram
6: Dataset + ML Models
7: Multi-Agent Control Tower
8: Prototype Demo Screenshots
9: Business Impact + ROI
10: Risks, Ethics, Future Improvements
11: Q&A

By: Cheung KH

Tools: Python • Scikit-learn • Pandas • Gradio • Hugging Face

2: Business Problem

Key Problems

- **Delivery Failures**
  Lead to re-delivery costs and poor customer experience

- **Unexpected Vehicle Breakdowns**
  Disrupt routes and cause delays

- **Uncertain Demand & Load Distribution**
  Leads to under-utilized / overloaded resources

2: Business Problem

Current Limitation

***Delivery Failures**

Most decisions are reactive (handle problems after they occur)

***Opportunity**

Can we predict risks BEFORE dispatch and take preventive action?

See full details: GitHub URL

3: Selected 5 Problems

Core Components

- **Demand Forecasting**
  Predict parcel volume to support resource planning
- **Delivery Failure Prediction**
  Identify high-risk deliveries before dispatch
- **Predictive Maintenance**
  Detect vehicles with high breakdown risk
- **Customer Support (RAG-lite)**
  Provide operational guidance based on system conditions
- **Multi-Agent Control Tower**
  Coordinate all components into unified decision system

3: Selected 5 Problems

Design Strategy

4 predictive/support components → 1 coordinating Control Tower

4: Solution Overview

AI Control Tower Approach

Use predictive models + multi-agent coordination
to enable proactive decision-making before dispatch

How the System Works

Operational Inputs
→ ML Predictions
→ Multi-Agent Coordination
→ Control Tower Decisions
→ Recommendations

4: Solution Overview

Key Idea

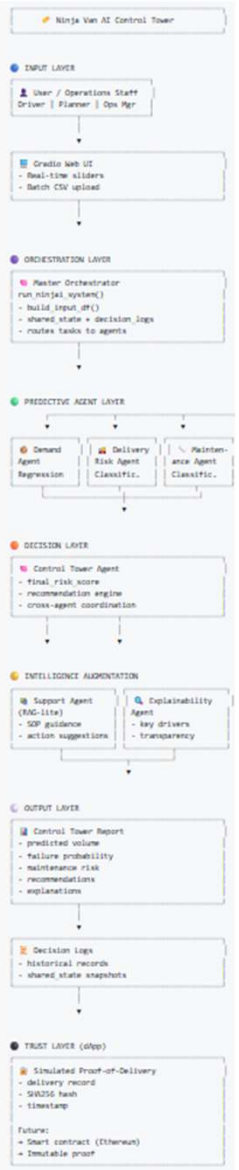
- Each component solves a **specific problem**
- Control Tower **combines all insights**
- Outputs are **actionable, not just predictions**

Outcome

Shift from reactive operations → Proactive, data-driven planning

5: Architecture Diagram

Multi-Agent Control Tower Architecture



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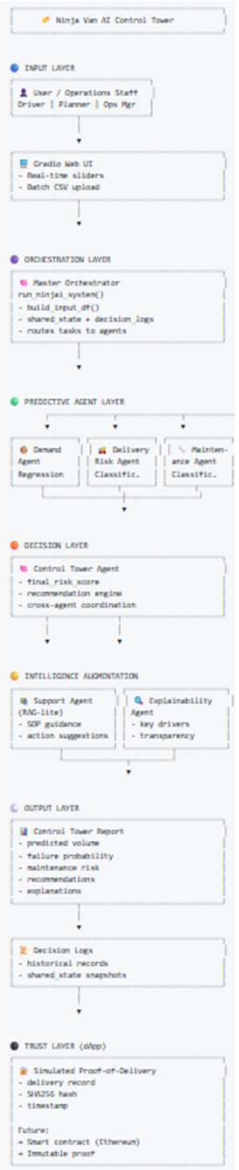

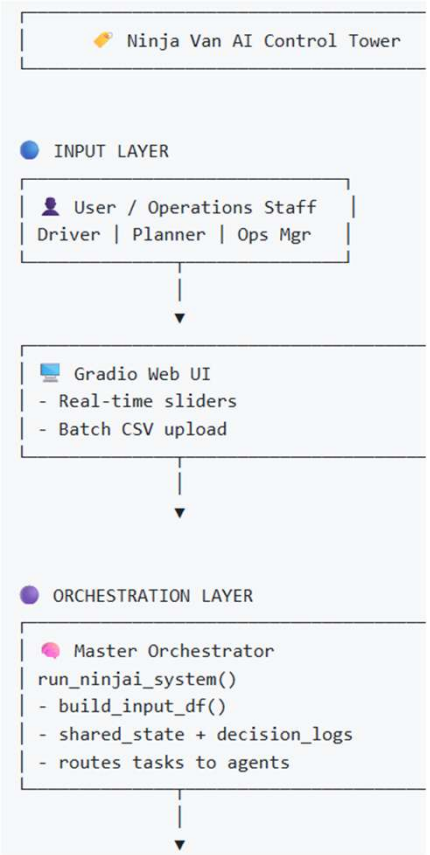

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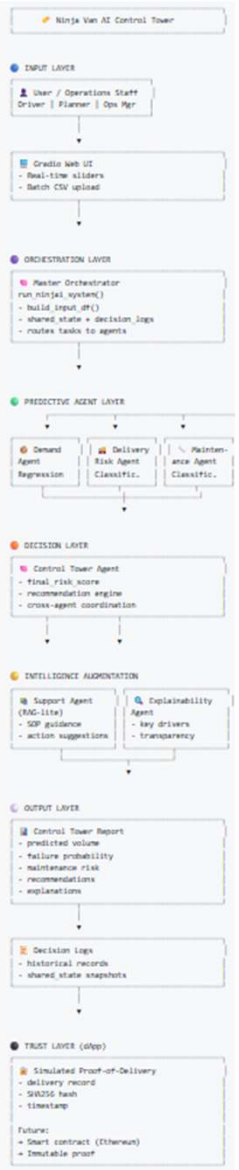

Multi-Agent Control Tower Architecture

```
Ninja Van AI Control Tower

INPUT LAYER
User / Operations Staff
Driver | Planner | Ops Mgr
        |
        ▼
Gradio Web UI
- Real-time sliders
- Batch CSV upload
        |
        ▼
ORCHESTRATION LAYER
Master Orchestrator
run_ninjai_system()
- build_input_df()
- shared_state + decision_logs
- routes tasks to agents
        |
        ▼
PREDICTIVE AGENT LAYER
Demand Agent Regression | Delivery Risk Agent Classific. | Mainten-ance Agent Classific.
        |
        ▼
DECISION LAYER
Control Tower Agent
- final_risk_score
- recommendation engine
- cross-agent coordination
        |
        ▼
INTELLIGENCE AUGMENTATION
Support Agent (RAG-lite)
- SOP guidance
- action suggestions
Explainability Agent
- key drivers
- transparency
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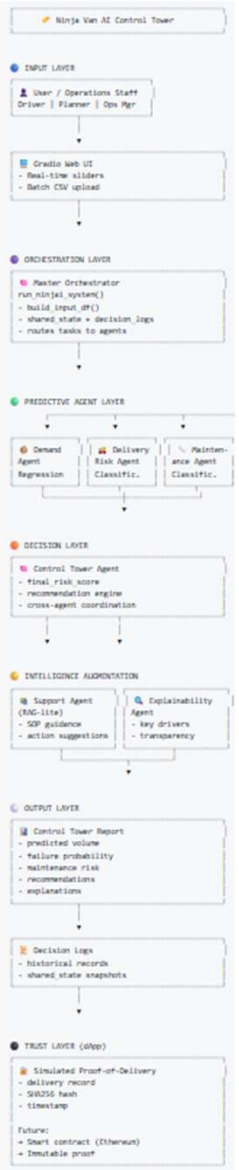

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OUTPUT LAYER
Control Tower Report
- predicted volume
- failure probability
- maintenance risk
- recommendations
- explanations
Decision Logs
- historical records
- shared_state snapshots

TRUST LAYER (dApp)
Simulated Proof-of-Delivery
- delivery record
- SHA256 hash
- timestamp
Future:
→ Smart contract (Ethereum)
→ Immutable proof
```

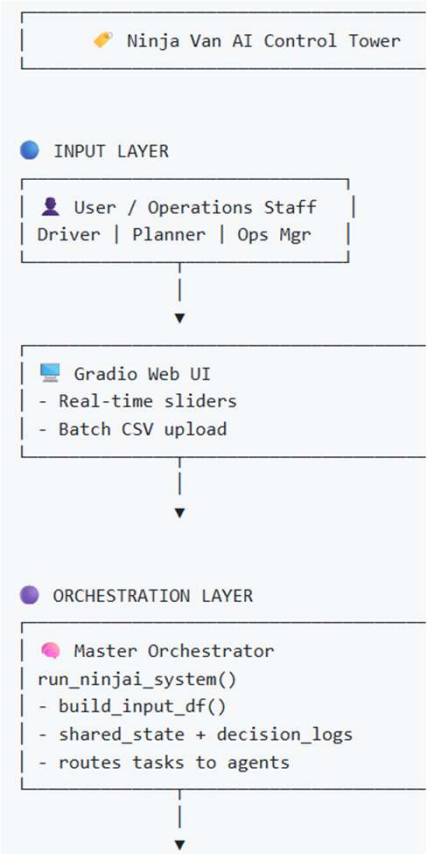

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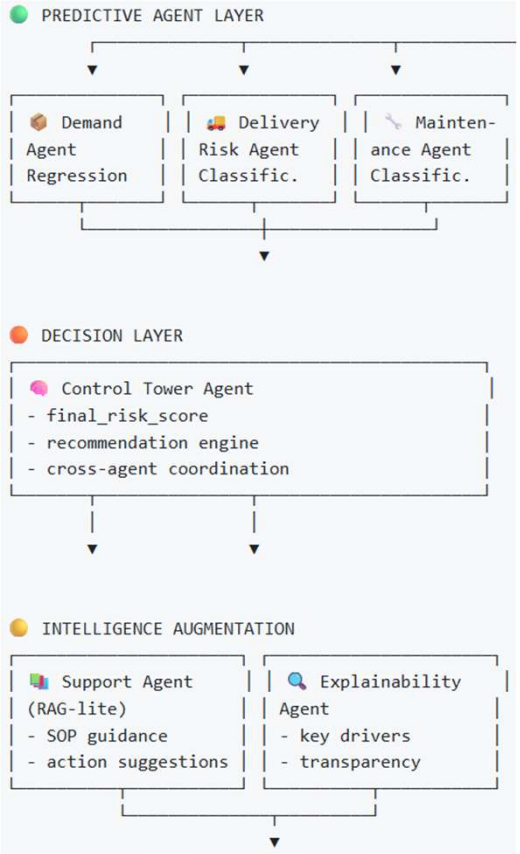

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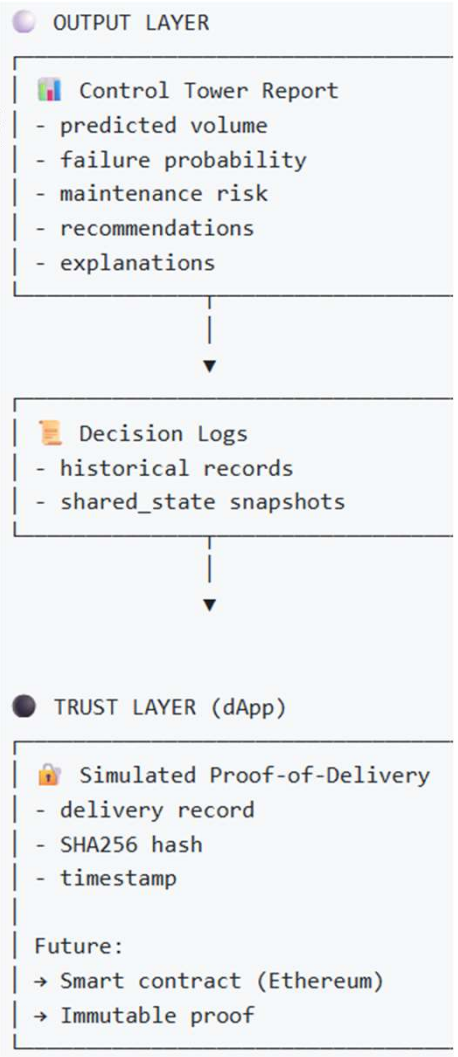

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5: Architecture Diagram

System Flow (short version)

Input → Prediction → Coordination → Decision → Output

5: Architecture Diagram

Key Components

- **Predictive Agents**
  Demand | Delivery Risk | Maintenance

- **Support & Explainability**
  RAG-lite guidance + explanation of predictions

- **Control Tower Agent**
  Combines all outputs into coordinated decisions

5: Architecture Diagram

Design Principle

Modular + extensible system architecture

6: Dataset + ML Models

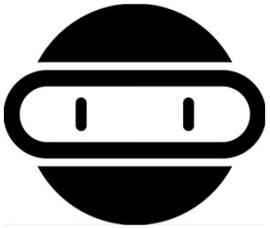
Dataset (Simulated)

- Synthetic logistics dataset (no real data available)
- Designed to mimic real operational conditions

Key Features

- distance_km, num_stops
- traffic_level, hub_congestion, weather_rain
- vehicle_mileage_km, days_since_service
- driver_experience, current_load

6: Dataset + ML Models



Model Selection

- **Demand Forecasting** → Regression model
- **Delivery Failure** → Classification model
- **Maintenance Risk** → Classification model

6: Dataset + ML Models

Justification

- Models match problem type (regression vs classification)
- Lightweight and fast → suitable for real-time use
- Easy to interpret and deploy

Design Choice

Controlled noise added to simulate real-world uncertainty

7: Multi-Agent Control Tower

<u>Agent-Based System Design</u>

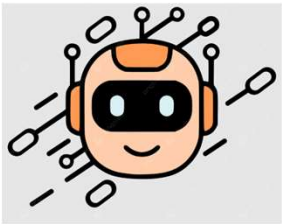
Multiple specialized agents → coordinated by a Control Tower

7: Multi-Agent Control Tower

Agent Roles

- 📦 **Demand Agent** → predicts parcel volume
- 🚚 **Delivery Risk Agent** → predicts failure probability
- 🔧 **Maintenance Agent** → predicts breakdown risk
- 📚 **Support Agent (RAG-lite)** → provides operational guidance
- 🔍 **Explainability Agent** → highlights key risk factors

7: Multi-Agent Control Tower



Control Tower Agent

- Combines outputs from all agents
- Computes overall risk score
- Generates coordinated recommendations

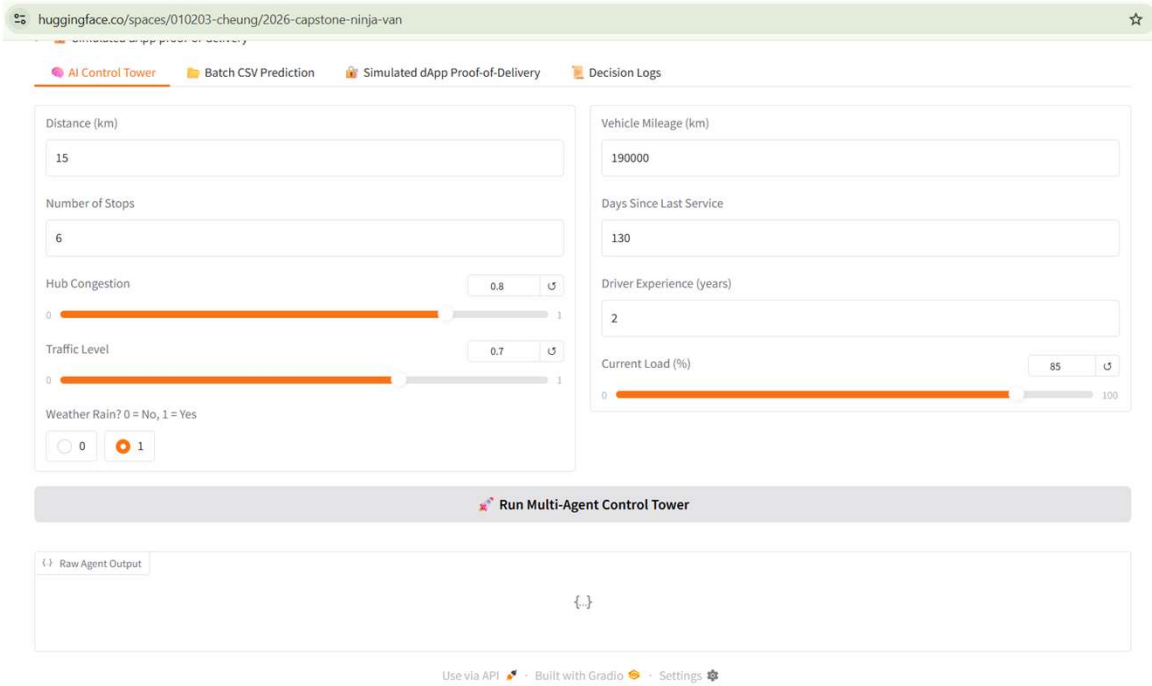
Key Advantage

Not just predictions → integrated decision-making system

8: Prototype Demo Screenshots

Interactive AI Control Tower (Live Demo)

- Real-time prediction interface
- Batch CSV upload
- Table-style output



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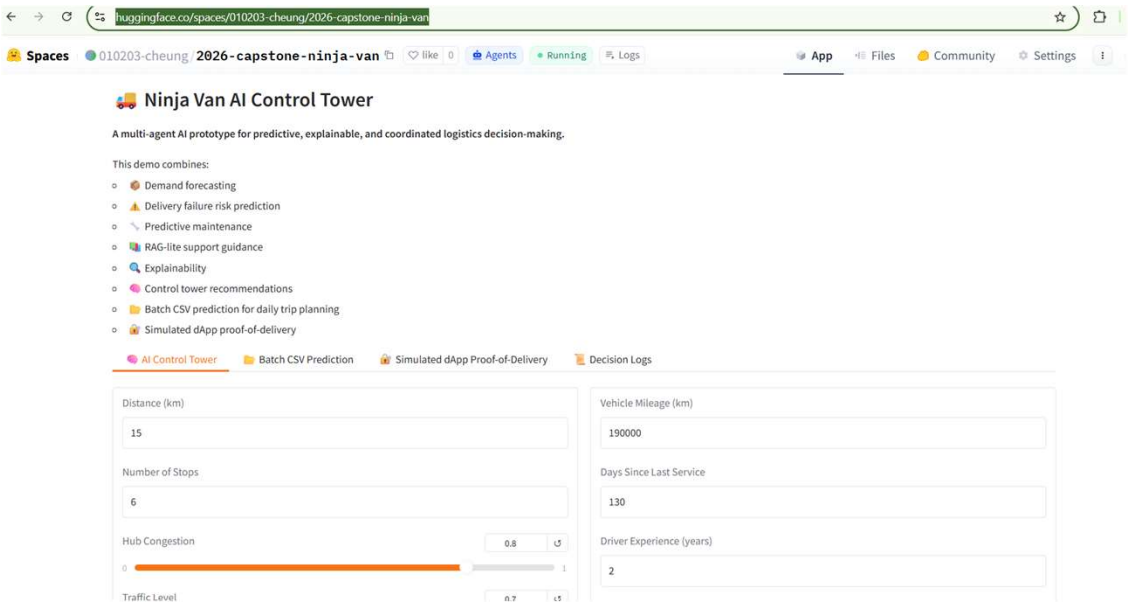

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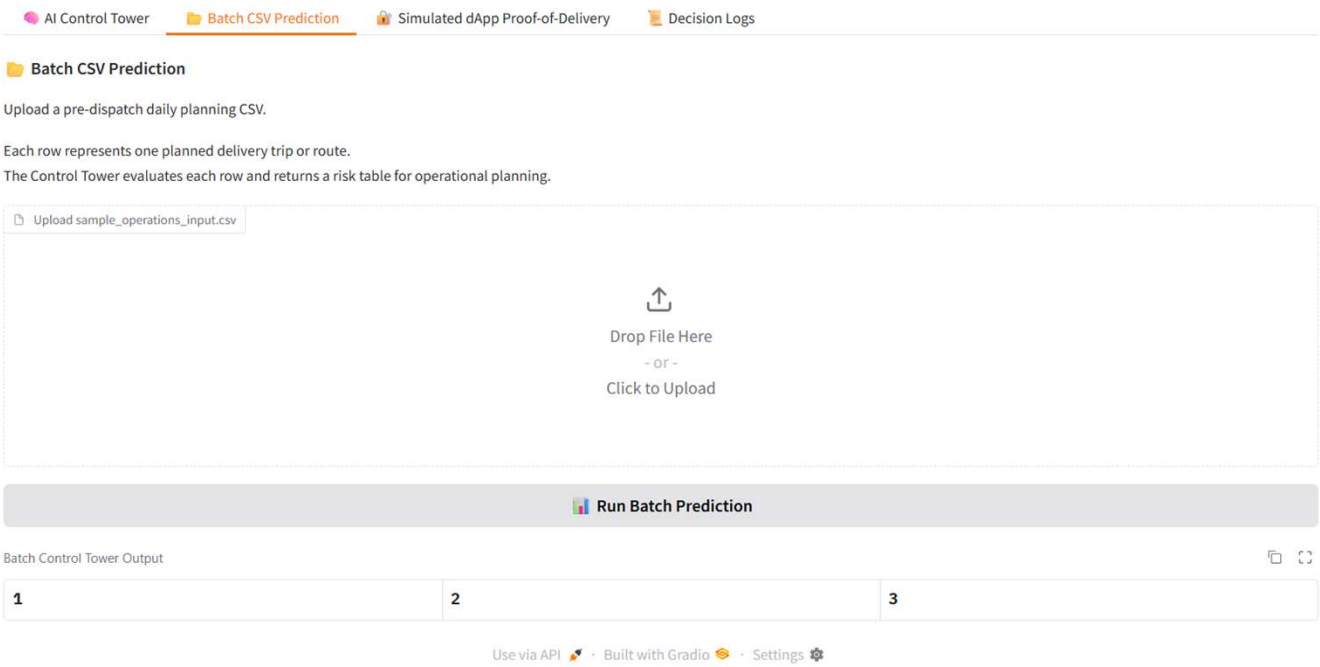

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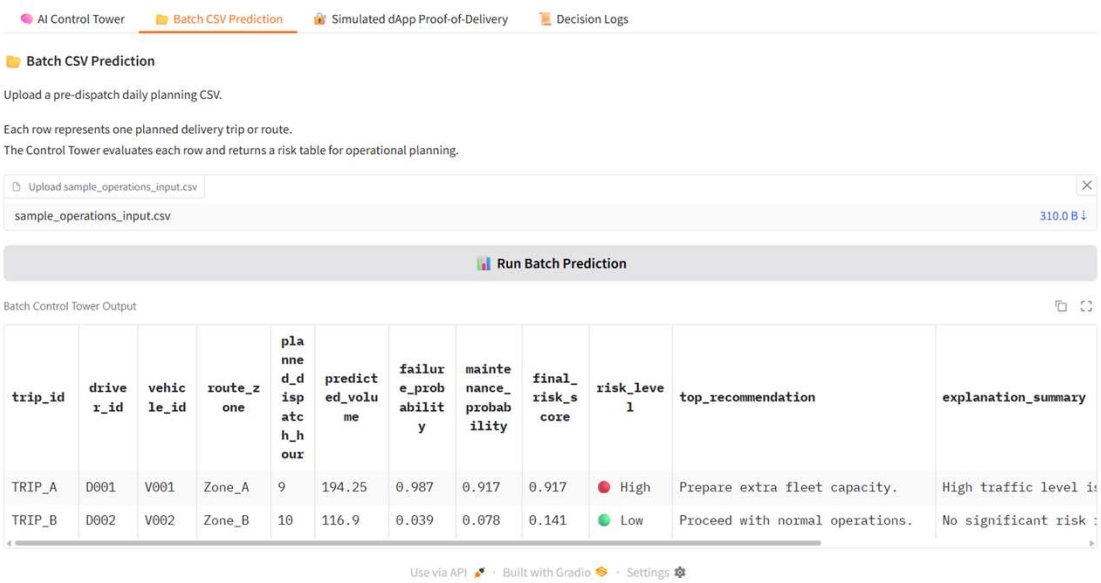


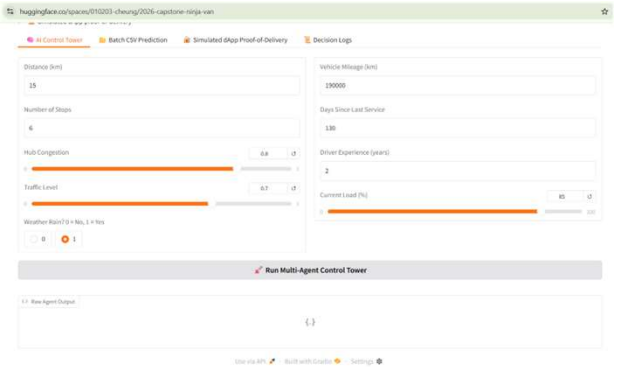
trip_id	driver_id	vehicle_id	route_zone	planned_dispatch_hour	predicted_volume	failure_probability	maintenance_probability	final_risk_score	risk_level	top_recommendation	explanation_summary
TRIP_A	D001	V001	Zone_A	9	194.25	0.987	0.917	0.917	High	Prepare extra fleet capacity.	High traffic level is
TRIP_B	D002	V002	Zone_B	10	116.9	0.039	0.078	0.141	Low	Proceed with normal operations.	No significant risk

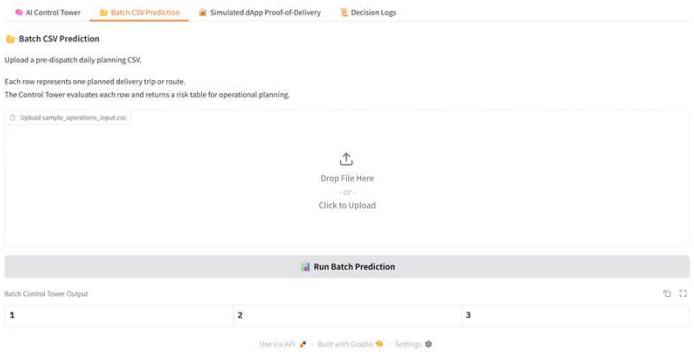
8: Prototype Demo Screenshots

Deployment Setup

- 🧠 Models trained in Google Colab
- 📦 Serialized using Joblib
- 🌐 Deployed via Gradio on Hugging Face Spaces

8: Prototype Demo Screenshots





<u>System Capabilities</u>

- ✅ Real-time scenario simulation (single trip)
- 📂 Batch operational planning (CSV upload)
- 📊 Risk scoring + recommendations
- 🔍 Explainability output
- 📚 RAG-lite support guidance

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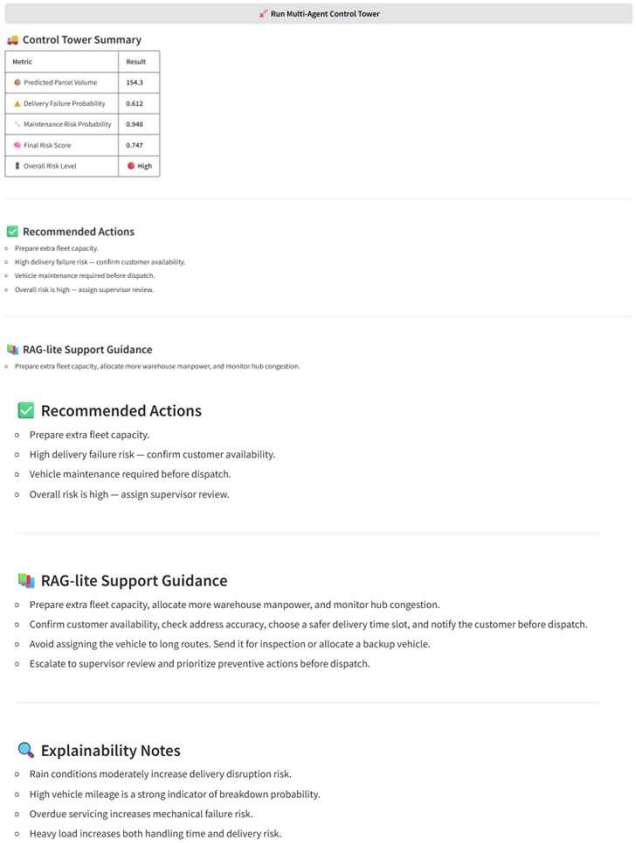

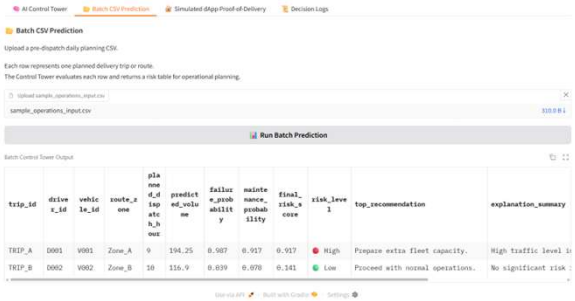

9: Business Impact + ROI

Value Drivers

- 🚚 Reduce failed deliveries (pre-dispatch intervention)
- 🔧 Prevent vehicle breakdowns (predictive maintenance)
- 📦 Optimize fleet & manpower allocation (demand forecasting)

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/roi_estimation.md

9: Business Impact + ROI

Estimated Cost Savings (Illustrative)

Component	Monthly Savings
Delivery failure reduction	~$30,000
Maintenance optimization	~$6,000
Total Estimated Benefit	**~$36,000/month**

Investment vs Return

- Estimated implementation cost: **$50K – $100K**
- Estimated payback period: **~2–3 months**

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/roi_estimation.md

9: Business Impact + ROI

Strategic Impact

- ✅ Improved customer satisfaction
- ✅ Reduced operational disruptions
- ✅ Shift to proactive decision-making
- ✅ Scalable across regions

Note

- Based on simulated data and assumptions
- Actual ROI depends on real-world data and adoption

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/roi_estimation.md

10: Risks, Ethics, Future Improvements

Key Risks

- 📊 **Model limitations**
  >>> Trained on simulated data → may differ in real deployment

- 🎯 **Prediction errors**
  >>> False positives → unnecessary actions
  >>> False negatives → missed intervention

- 🔄 **Operational dependency**
  >>> Risk of over-reliance on AI outputs

- 📥 **Data quality risk**
  >>> Inaccurate inputs → unreliable predictions

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/risk_ethics.md

10: Risks, Ethics, Future Improvements

Ethical Safeguards

- **Human-in-the-loop**
  >>> AI supports decisions, does not replace them
- **Explainability**
  >>> Key risk factors (traffic, weather, vehicle) are surfaced
- **Transparency & logging**
  >>> Decision logs ensure traceability
- **Privacy awareness**
  >>> Prototype avoids personal data; future systems require safeguards

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/risk_ethics.md

10: Risks, Ethics, Future Improvements

Current System (RAG-lite)

- Rule-based knowledge base (SUPPORT_KB)
- Maps system conditions → predefined guidance
- Fast and deterministic

Limitation

- Cannot scale to real operational knowledge
- No semantic retrieval or dynamic responses

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/rag_upgrade_path.md

10: Risks, Ethics, Future Improvements

🚀 Full RAG Upgrade (Future)

```
System Event / Query
→ Embedding
→ Vector DB (SOPs, FAQs)
→ Retrieval
→ LLM-generated guidance
```

What This Enables

- 📂 Dynamic retrieval from SOPs & policies
- 🔍 Context-aware operational guidance
- 🔄 Continuous knowledge updates (no retraining)

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/rag_upgrade_path.md

10: Risks, Ethics, Future Improvements

Example Stack

- Embeddings: sentence-transformers
- Vector DB: FAISS / ChromaDB
- Framework: LangChain / LlamaIndex

🧠 Key Upgrade

```
From rule-based guidance
→ to dynamic, context-aware knowledge retrieval
```

See full details: https://github.com/010203cheung/ninja-van-ai-control-tower/blob/main/docs/rag_upgrade_path.md

Q&A